

Homework 5

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Physics 16

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Question 1. Purcell chapter 4, exercise 4.19 (Synchrotron current.) In a 6 giga-electron-volt ($1 \text{ GeV} = 10^9 \text{ eV}$) electron synchrotron, electrons travel around the machine in an approximately circular path 240 meters long. It is normal to have about 10^{11} electrons circling on this path during a cycle of acceleration. The speed of the electrons is practically that of light. What is the current?

Solution. Recall that current is

$$I = \frac{Q}{t}.$$

Since the 240 m is the total path length (circumference), the time to complete one revolution is

$$T = \frac{d}{v} = \frac{240 \text{ m}}{3 \times 10^8 \text{ m/s}} = 8 \times 10^{-7} \text{ s}.$$

The current is then

$$I = \frac{ne}{T} = \frac{10^{11}(1.6 \times 10^{-19} \text{ C})}{8 \times 10^{-7} \text{ s}} = \frac{1.6 \times 10^{-8}}{8 \times 10^{-7}} = \boxed{0.02 \text{ A}}.$$

□

Question 2. Purcell chapter 4, exercise 4.29 (Keeping the same resistance.) In the circuit shown in Fig. 4.47, if R_0 is given, what value must R_1 have in order that the input resistance between the terminals shall be equal to R_0 ?

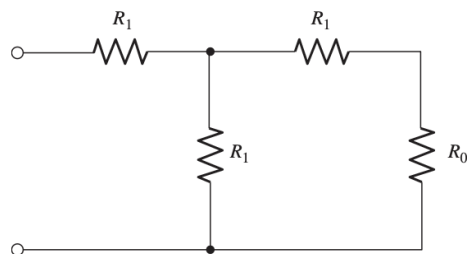


Figure 4.47.

Solution. Given the terminal net resistance is R_0 , by resistor rules, we get

$$R_0 = R_1 + \left(\frac{1}{R_1} + \frac{1}{R_1 + R_0} \right)^{-1} = R_1 + \frac{R_1(R_1 + R_0)}{2R_1 + R_0}.$$

We further reduce the equation as such:

$$R_0(2R_1 + R_0) = R_1(2R_1 + R_0) + R_1(R_1 + R_0)$$

$$2R_0R_1 + R_0^2 = 2R_1^2 + R_0R_1 + R_1^2 + R_0R_1 = 3R_1^2 + 2R_0R_1.$$

The $2R_0R_1$ terms cancel, leaving

$$R_0^2 = 3R_1^2 \implies R_0 = \sqrt{3} R_1.$$

$$R_1 = \frac{R_0}{\sqrt{3}}.$$

□

Question 3. Purcell chapter 4, exercise 4.38 (Two light bulbs.)

- (a) Two light bulbs are connected in parallel, and then connected to a battery, as shown in Fig. 4.51(a). You observe that bulb 1 is twice as bright as bulb 2. Assuming that the brightness of a bulb is proportional to the power dissipated in the bulb's resistor, which bulb's resistor is larger, and by what factor?
- (b) The bulbs are now connected in series, as shown in Fig. 4.51(b). Which bulb is brighter, and by what factor? How bright is each bulb compared with bulb 1 in part (a)?

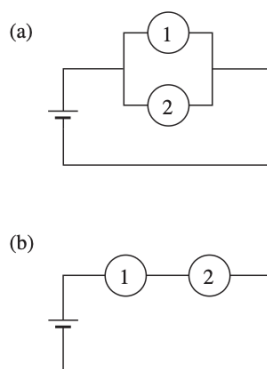


Figure 4.51.

Solution. For part (a) the voltage across each bulb is the same. Power dissipated is

$$P = \frac{V^2}{R},$$

so the brighter bulb (more power) has *lower* resistance. If bulb 1 is twice as bright as bulb 2, then $R_2 = 2R_1$. For part (b), the current through each bulb is the same. Power dissipated is

$$P = I^2 R,$$

so the brighter bulb (more power) has *higher* resistance. Hence, $R_1 = 2R_2$. $\square$

Question 4. Purcell chapter 4, exercise 4.47 (Discharging a capacitor.) For a capacitor C discharging through a resistor R (worked out in Purcell Section 4.11), show that the total energy dissipated in the resistor agrees with the energy originally stored in the capacitor.

Suppose someone objects that the capacitor is never *really* discharged because Q becomes zero only for $t = \infty$. How would you counter this objection? You might find out how long it would take the charge to be reduced to one electron, with some reasonable assumptions.

Solution. For an RC circuit, the charge on the capacitor is $Q(t) = Q_0 e^{-t/RC}$, so the current is

$$I(t) = \dot{Q}(t) = \frac{Q_0}{RC} e^{-t/RC}.$$

The total energy dissipated in the resistor is

$$E = \int_0^\infty I^2 R dt = \int_0^\infty \frac{Q_0^2}{RC^2} e^{-2t/RC} dt = \frac{Q_0^2}{RC^2} \left[\frac{-RC}{2} e^{-2t/RC} \right]_0^\infty = \frac{Q_0^2}{RC^2} \cdot \frac{RC}{2} = \frac{Q_0^2}{2C},$$

which is exactly the initial energy of the capacitor, $U = Q_0^2/2C$. To counter the objection mentioned in the latter part, we set $Q(t)$ equal to one electron charge e :

$$e = Q_0 e^{-t/RC} \implies t = RC \ln\left(\frac{Q_0}{e}\right).$$

With reasonable values (say $C = 1 \mu\text{F}$, $R = 1 \text{ M}\Omega$, charged to $V_0 = 10 \text{ V}$ so $Q_0 = CV_0 = 10 \mu\text{C}$):

$$t = (1 \mu\text{F})(1 \text{ M}\Omega) \ln\left(\frac{10 \times 10^{-6}}{1.6 \times 10^{-19}}\right) \approx 31.8 \text{ s}.$$

This is completely finite and quite short, so the capacitor is effectively fully discharged long before $t = \infty$. $\square$

Question 5. Suppose a wire of resistance R could be stretched uniformly until it was twice its original length (while keeping its mass density constant). What would happen to the resistance?

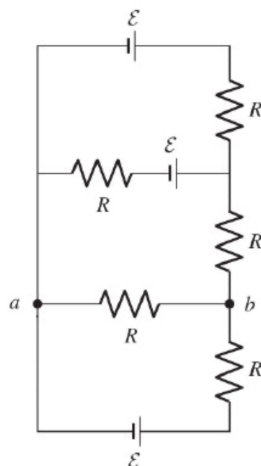
Solution. Recall that the resistance of a wire is $R = \rho L/A$. When the wire is stretched to twice its length, the volume is fixed, hence when length doubles, the cross-sectional area must halve. Hence, the new resistance is

$$R' = \rho \frac{2L}{A/2} = 4 \frac{\rho L}{A} = \boxed{4R.}$$

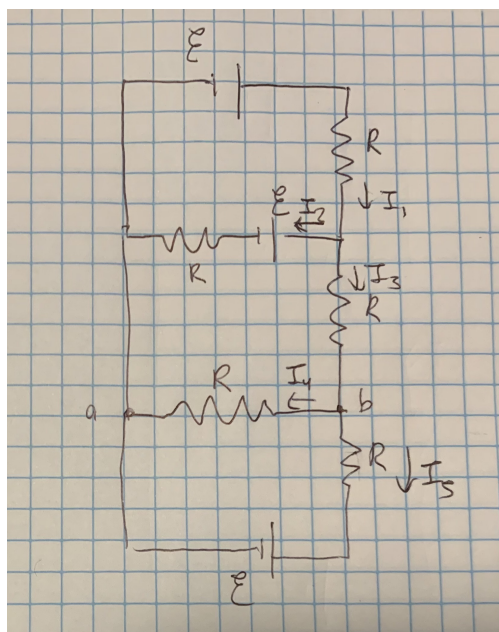
The resistance increases by a factor of 4.

□

Question 6. What is the potential difference between points a and b in the circuit shown in the figure below?



Solution. We draw the currents as such. We define the set rules:



Junction rules:

$$I_1 = I_2 + I_3, \quad I_3 = I_4 + I_5.$$

Loop rules:

$$\mathcal{E} - I_1 R - \mathcal{E} - I_2 R = 0 \implies I_1 = -I_2,$$

$$\mathcal{E} - I_1 R - I_3 R - I_4 R = 0 \implies I_1 + I_3 + I_4 = \frac{\mathcal{E}}{R},$$

$$\mathcal{E} - I_1 R - I_3 R - I_5 R - \mathcal{E} = 0 \implies I_1 + I_3 + I_5 = 0.$$

From $I_1 = -I_2$ and the first junction rule,

$$-I_2 = I_2 + I_3 \implies I_3 = -2I_2.$$

Substituting $I_1 = -I_2$ and $I_3 = -2I_2$ into the second loop equation:

$$-I_2 - 2I_2 + I_4 = \frac{\mathcal{E}}{R} \implies I_4 = \frac{\mathcal{E}}{R} + 3I_2.$$

From the second junction rule,

$$I_5 = I_3 - I_4 = -2I_2 - \frac{\mathcal{E}}{R} - 3I_2 = -\frac{\mathcal{E}}{R} - 5I_2.$$

Substituting into the third loop equation gives

$$-I_2 - 2I_2 + \left(-\frac{\mathcal{E}}{R} - 5I_2\right) = 0 \implies -8I_2 = \frac{\mathcal{E}}{R} \implies I_2 = -\frac{\mathcal{E}}{8R}.$$

Therefore,

$$I_1 = \frac{\mathcal{E}}{8R}, \quad I_3 = \frac{\mathcal{E}}{4R}, \quad I_4 = \frac{5\mathcal{E}}{8R}, \quad I_5 = -\frac{3\mathcal{E}}{8R}.$$

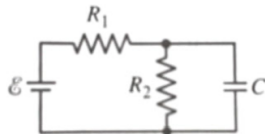
The potential difference between a and b along the branch carrying I_4 is:

$$\boxed{V_a - V_b = I_4 R = \frac{5\mathcal{E}}{8}.$$

□

Question 7.

- (a) Apply Kirchhoff's rules for the circuit shown to find a differential equation for the charge on the capacitor as a function of time.



- (b) Come up with a solution $Q(t)$ for the charge on the capacitor as a function of time, with $Q(t = 0) = 0$.

Solution. In part (a), we use Kirchhoff's junction rule at the node where R_1 , R_2 , and C meet

$$I_1 = I_2 + I_C,$$

where I_1 is the current through R_1 , I_2 is the current through R_2 , and $I_C = dQ/dt$ is the current into the capacitor. Since R_2 and C are in parallel, they share the same voltage Q/C , so $I_2 = Q/(R_2C)$. Thus:

$$I_1 = \frac{dQ}{dt} + \frac{Q}{R_2C}.$$

Applying Kirchhoff's voltage law around the outer loop (through $\mathcal{E}$, R_1 , and C) gives

$$\mathcal{E} - I_1 R_1 - \frac{Q}{C} = 0.$$

$$\mathcal{E} - R_1 \left(\frac{dQ}{dt} + \frac{Q}{R_2C} \right) - \frac{Q}{C} = 0.$$

$$\boxed{R_1 \frac{dQ}{dt} + \frac{Q}{C} \cdot \frac{R_1 + R_2}{R_2} = \mathcal{E}.}$$

For part (b), let us rearrange this in a generalized form $\dot{Q} + Q/\tau = \frac{\mathcal{E}}{R_1}$, where

$$\tau = \frac{R_1 R_2 C}{R_1 + R_2}.$$

The steady-state charge on the capacitor (as $t \rightarrow \infty$, $dQ/dt = 0$) is

$$Q_{\max} = \frac{\mathcal{E} R_2 C}{R_1 + R_2}.$$

The general solution is

$$Q(t) = Q_{\max} + A e^{-t/\tau}.$$

Applying the initial condition $Q(0) = 0$ gives

$$0 = Q_{\max} + A \implies A = -Q_{\max}.$$

Therefore,

$$Q(t) = \frac{\mathcal{E} R_2 C}{R_1 + R_2} \left(1 - e^{-t(R_1 + R_2)/(R_1 R_2 C)}\right).$$

□

Question 8. Purcell chapter 6, exercise 6.30 (High-energy proton in space.) A proton with kinetic energy 10^{16} eV ($\gamma = 10^7$) is moving perpendicular to the interstellar magnetic field, which in that region of the galaxy has a strength 3×10^{-6} gauss. What is the radius of curvature of its path and how long does it take to complete one revolution? (This is a relativistic problem. You may use the results from Purcell Exercise 6.29.)

Solution. Using the previous work in Exercise 6.29, and assuming the proton is essentially moving at the speed of light due to the extremely large Lorentz factor, we get that

$$R = \frac{p}{qB} = \frac{\gamma mv}{qB} = \frac{10^7(1.67 \times 10^{-27} \text{ kg})(3 \times 10^8 \text{ m/s})}{(1.6 \times 10^{-19} \text{ C})(3 \times 10^{-10} \text{ T})} \approx 1.04 \times 10^{17} \text{ m}$$

Then, the period of one revolution is

$$T = \frac{2\pi R}{v} = \frac{2\pi(1.04 \times 10^{17} \text{ m})}{3 \times 10^8 \text{ m/s}} \approx \boxed{2.2 \times 10^9 \text{ s} \approx 69 \text{ years.}}$$

□